

CLOUD SECURITY MONITORING WITH MICROSOFT SENTINEL

Azure SIEM/SOAR Proof of Concept | Data Ingestion, Validation & Security Visibility

INDIVIDUAL CONTRIBUTION Data ingestion and validation: Azure Activity Logs + Microsoft Defender for Cloud, with KQL verification.	TEAM PoC 10-day Azure project covering monitoring, threat detection, automation, hunting, dashboards, testing, and cost analysis.
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THE SECURITY CHALLENGE

The Proof of Concept evaluated Microsoft Sentinel as a cloud-native SIEM/SOAR platform for centralising Azure security telemetry, improving threat detection, automating response workflows, supporting proactive threat hunting, and providing operational reporting. A strong data foundation was essential: security events had to reach Log Analytics and remain queryable before detection or response could work reliably.

MY CONTRIBUTION: BUILDING & VALIDATING THE DATA FOUNDATION

Connect Integrated Azure Activity Logs to provide visibility into Azure management-plane activity.	Enrich Integrated Microsoft Defender for Cloud to add security alerts, recommendations, compliance findings, vulnerability information, and resource assessments.	Validate Confirmed that telemetry was reaching the Log Analytics Workspace and was accessible in Microsoft Sentinel.	Query Used KQL to verify ingestion and review Azure administrative and security activity.
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SOLUTION ARCHITECTURE

Azure Activity Logs	Microsoft Defender for Cloud
Management-plane activity Role assignments • resource changes • policy changes • administrative operations	Security alerts • recommendations • compliance findings • vulnerability & resource assessments
↓	↓
Log Analytics Workspace Central telemetry repository • KQL querying • analytics foundation	
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Microsoft Sentinel Analytics • incidents • hunting • workbooks • automation	

WHY DATA INGESTION MATTERED

Azure Activity Logs supplied the administrative audit trail needed to monitor resource creation/deletion, role assignments, policy changes, virtual machine actions, security configuration changes, and other privileged operations. Defender for Cloud added posture and threat context. Together, the two sources created the telemetry foundation for analytics, investigation, and threat detection.

KQL VALIDATION & TEAM SECURITY PIPELINE

Proving that telemetry was searchable - then validating the broader detection and response workflow

KQL VALIDATION EXAMPLES

Verify Azure Activity ingestion	
AzureActivity where TimeGenerated > ago(24h) summarize Events=count()	Purpose A non-zero count confirms Azure Activity events are flowing into Sentinel.
Review recent administrative actions	
AzureActivity where TimeGenerated > ago(24h) project TimeGenerated, Caller, OperationName, ActivityStatus sort by TimeGenerated desc	Purpose Provides a recent audit view of who performed Azure administrative actions and their status.
Verify Defender for Cloud alerts	
SecurityAlert where TimeGenerated > ago(7d) project TimeGenerated, AlertName, Severity, CompromisedEntity sort by TimeGenerated desc	Purpose Confirms Defender for Cloud security alerts are reaching Sentinel.

BROADER TEAM PoC: END-TO-END SIEM/SOAR PIPELINE



Validation Item	Actual Outcome	Status
Data Ingestion	Logs received	PASS
Analytics Rule	Alert generated	PASS
Incident Creation	Incident created	PASS
Playbook Trigger	Workflow executed	PASS
Notification Delivery	Notification sent	PASS
Workbook Visibility	Event displayed	PASS

The team used a controlled privilege-escalation scenario involving elevated Azure permissions outside approved business hours. The documented test results show successful ingestion, alert generation, incident creation, playbook execution, notification delivery, and workbook visibility.

SECURITY VALUE, COMPLIANCE & COST AWARENESS

Turning centralised telemetry into operational visibility while keeping ownership claims clear

WHAT MY WORK DEMONSTRATED

Cloud Telemetry Connected security-relevant Azure sources into a central monitoring architecture.	KQL Validation Verified that data was flowing and usable rather than assuming a connector status meant success.	Audit Visibility Made Azure administrative activity searchable for monitoring and investigation.	Security Context Combined activity telemetry with Defender for Cloud security and posture information.
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TEAM OUTCOMES

The completed PoC extended the data foundation into built-in and custom analytics, incident generation, Logic Apps automation, threat hunting, and a Sentinel Workbook for operational and executive visibility. The project report states that the complete monitoring pipeline functioned as intended and that all validation objectives were achieved.

Detection	Built-in and custom analytics transformed incoming telemetry into actionable alerts and incidents.
Automation	A Logic Apps playbook automatically created security notifications when qualifying incidents were generated.
Threat Hunting	KQL was used to proactively search telemetry for suspicious activity beyond triggered analytics rules.
Reporting	A Sentinel Workbook supported incident monitoring, alert analysis, security trends, operational visibility, and executive reporting.

COMPLIANCE & GOVERNANCE VALUE

The project materials frame centralised logging, searchable audit activity, security monitoring, and incident-response evidence as important to SOC 2 and ISO 27001 readiness. The dedicated Azure resource group also supported governance, access management, resource tracking, and cost monitoring. This portfolio case study presents those as team/project outcomes rather than claiming individual ownership of the full compliance implementation.

COST AWARENESS

£4.34 Actual cost at analysis	~£0.57/day Average daily spend	£24.61 Forecast for selected reporting period
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The report attributes the low PoC cost to limited ingestion volume and the controlled test environment, while using Azure Cost Management to illustrate how SIEM design should consider operational expenditure as well as technical capability.

PORTFOLIO POSITIONING

My contribution Azure Activity Logs • Defender for Cloud integration • data-ingestion validation • KQL verification • cloud visibility	Collaborative team delivery Analytics rules • privilege-escalation simulation • incidents • Logic Apps automation • threat hunting • workbook/dashboard • end-to-end testing
SKILLS DEMONSTRATED	Microsoft Sentinel • Azure Log Analytics • Azure Activity Logs • Microsoft Defender for Cloud • KQL • SIEM • Cloud Security Monitoring • Data Ingestion • Security Validation • Audit Visibility • Compliance Awareness

Portfolio case study | Uyo V. Alhassan. Individual contribution is separated from broader Cloud Security Cohort Team outcomes to preserve accurate ownership.